Matrix Theory, Voluntary Assignment 1

1 Instructions

In this assignment we will apply some concepts from Chapters 3 and 4 to a few problems. The main purpose is to provide feedback on your written proofs and solutions.

Note that the assignment is optional. You do not have to do it and it does not give any kind of bonus later in the course. It is intended to be an opportunity to use the theoretical tools of the course and get feedback on what you do. **If you want feedback hand in your solutions by the end of study week 4, as scanned pdf-files through the canvas page**. Make sure all your solutions are well structured, motivated and easy to follow.

If you have questions or run into problems you can go to the exercise sessions to get help (see course schedule) or send an email to the lecturer (`carl.olsson@math.lth.se`).

2 Span, determinants and some applications in optimization

Consider the set

$$\mathcal{C} = \{x \in \mathbb{R}^n;\ Ax \leq b\}, \tag{1}$$

where $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^{m\times 1}$. (Here "$\leq$" is a vector inequality meaning that each element of the vector to the left is smaller than the corresponding element of the vector to the right.) The set $\mathcal{C}$ is an intersection of half-spaces $a_i^T x \leq b_i$, $i = 1, ..., m$ where a_i^T is row i of A and b_i is element i of b. It is convex, meaning that if two points are in the set then so is the line segment between them, or in mathematical terms if

$$x, y \in \mathcal{C} \Rightarrow \lambda x + (1-\lambda)y \in \mathcal{C} \text{ for any } \lambda \in (0, 1). \tag{2}$$

Exercise 1. If $x, y \in \mathcal{C}$ and $z = \lambda x + (1-\lambda)y$ for some $\lambda \in (0, 1)$ show that

$$a_i^T z = b_i \Leftrightarrow \begin{cases} a_i^T x = b_i \\ a_i^T y = b_i \end{cases}. \tag{3}$$

A point $z \in \mathcal{C}$ is called an extreme point if there is no line segment in $\mathcal{C}$ that contains z in its interior, that is, if $z = \lambda x + (1-\lambda)y$ for some $\lambda \in (0, 1)$ and $x, y \in \mathcal{C}$ then $z = x = y$.

Exercise 2. Let $\mathcal{I}(z) = \{i; a_i^T z = b_i\}$, that is $\mathcal{I}(z)$ contains indices of the constrains where equality holds. Show that z is an extreme point if and only if the vectors a_i, $i \in \mathcal{I}(z)$ span $\mathbb{R}^n$.
(Hint: Consider the matrix $A_{\mathcal{I}(z)}$ containing the rows $a_i^T, i \in \mathcal{I}(z)$.)

Exercise 3. Suppose that $z = \lambda x + (1-\lambda)y$, with $x, y \in \mathcal{C}$ for some $\lambda \in (0, 1)$. Show that if $f(x) = r^T x$, that is f is linear, then

$$\begin{cases} f(x) \leq f(z) \\ f(y) \leq f(z) \end{cases} \Leftrightarrow f(x) = f(y) = f(z). \tag{4}$$

In the following we will say that the set $\mathcal{C}$ is bounded if it is non-empty and for each direction $v \in \mathbb{R}^n$ there is a row a_i^T in the matrix A such that $a_i^T v > 0$. (This technical condition means that the feasible set is bounded in the regular sense.)

Exercise 4. Suppose that we want to maximize the linear function $f(x) = r^T x$ over the bounded set $\mathcal{C}$. Show that the maximal value is attained in an extreme point.
(Hint: First show that if the maximum is attained in $z \in \mathcal{C}$ and the rows of $A_{\mathcal{I}(z)}$ do not span $\mathbb{R}^n$ then there is another point $z' \in \mathcal{C}$ such that $f(z') = f(z)$ and $\mathcal{I}(z) \subsetneq \mathcal{I}(z')$. Then use induction.)

Also find the largest value of $f(x)$ over $\mathcal{C}$ when

$$r = \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ -2 & -1 \\ 0 & -1 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}. \tag{5}$$

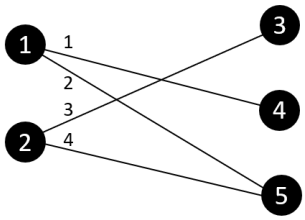
In the above example we solved a continuous optimization problem by reducing it to an optimization problem over a finite number of points. The latter problem is easily solved since the number of extreme points is very low. In in many large scale combinatorial problems the number of points that need to be examined is far too large for this to be feasible. In this case it is often desirable to do the opposite. That is, from a problem with a discrete (but large) number of points we want to formulate a continuous optimization problem, that has the same solution as the original one. In the following exercises we present one example of such a problem.

A matrix for which the determinants of all square sub-matrices are either $0, -1$ or 1 is called totally unimodular.

Exercise 5. Show that if $\mathcal{C}$ is bounded, A is totally unimodular and b has integer elements then the problem of maximizing $f(x) = r^T x$ over $\mathcal{C}$ has an integer solution x.

For an undirected graph we can form an incidence matrix $G = (g_{ij})$. The rows of G correspond to nodes of the graph and the columns correspond to edges. An element g_{ij} equals 1 if node i is connected to edge j and 0 otherwise. Since each edge connects two nodes, each column will contain two elements that are 1. The sum over a row is the degree of the corresponding node (that is, number of edges the node is connected to).

In a bipartite graph we can divide the nodes into two groups such that the graph has no edges within any of the two groups. Thus, after reordering the rows/nodes the matrix G can be written $G = \begin{bmatrix} G' \\ G'' \end{bmatrix}$, where the columns of G' and G'' contain one element that is 1.



$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

Figure 1: An example of a bipartite graph and its incidence matrix.

Exercise 6. Show that if the graph is bipartite then G is totally unimodular by using induction over the size of the sub-matrices. (Hint: Consider three cases, either the submatrix has a column of all zeros, or a column with a single 1 or all columns have two elements that are 1.)

In a bipartite matching or assignment problem we seek to uniquely assign nodes in one group to corresponding nodes in the other group, by turning edges of the graph on or off. Let x be a vector whose elements only take

binary values. If element $x_i = 1$ we use edge i to match the two nodes that are connected by edge i otherwise $x_i = 0$. The constraint $Gx \leq e$, where e is a vector of all ones, ensures that each node is only connected to one edge which makes the matching of nodes one-to-one. The use of edge i comes with a reward r_i and we which to find the matching that maximizes the total reward $\sum_i r_i x_i = r^T x$.

We thus get the optimization problem

$$\begin{aligned} \text{maximize} \quad & r^T x, && (6)\\ \text{such that} \quad & Gx \leq e, && (7)\\ \text{and} \quad & x \in \mathbb{B}^n. && (8) \end{aligned}$$

Here $\mathbb{B}^n$ is the set of n-dimensional binary vectors.

The main difficulty here is the binary constraint. The following exercise shows that it can be replaced by the constraint $x_i \geq 0$, without changing the result. The problem can then be solved much more efficiently using local search methods.

Exercise 7. Let

$$\mathcal{S} = \{x \in \mathbb{R}^n;\ Gx \leq e, x \in \mathbb{B}^n\} \tag{9}$$

and

$$\mathcal{C} = \{x \in \mathbb{R}^n;\ Gx \leq e, x \geq 0\} \tag{10}$$

Show that $\mathcal{S} \subset \mathcal{C}$ and that if the graph is bipartite then the extreme points of $\mathcal{C}$ are binary.
(Hint: Consider the matrix $A = \begin{bmatrix} G \\ -I \end{bmatrix}$.)